

Arrhythmias

prepared by

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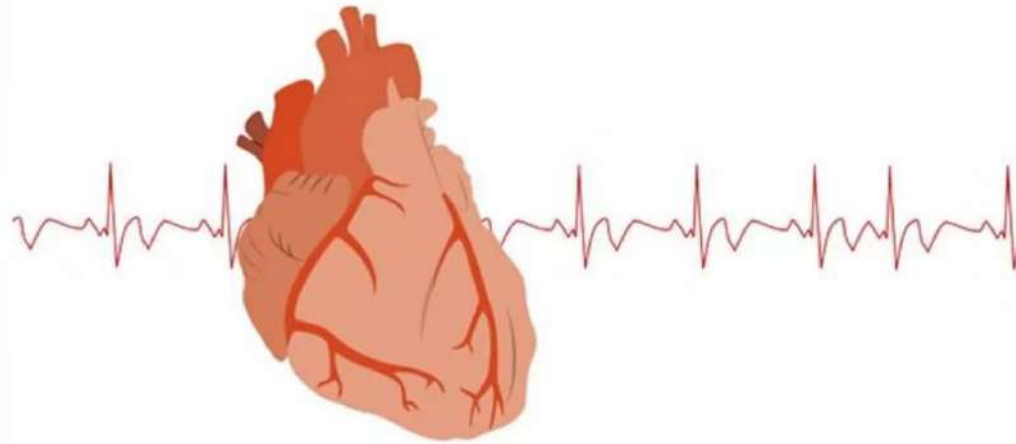
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Intensive Care

Arrhythmias

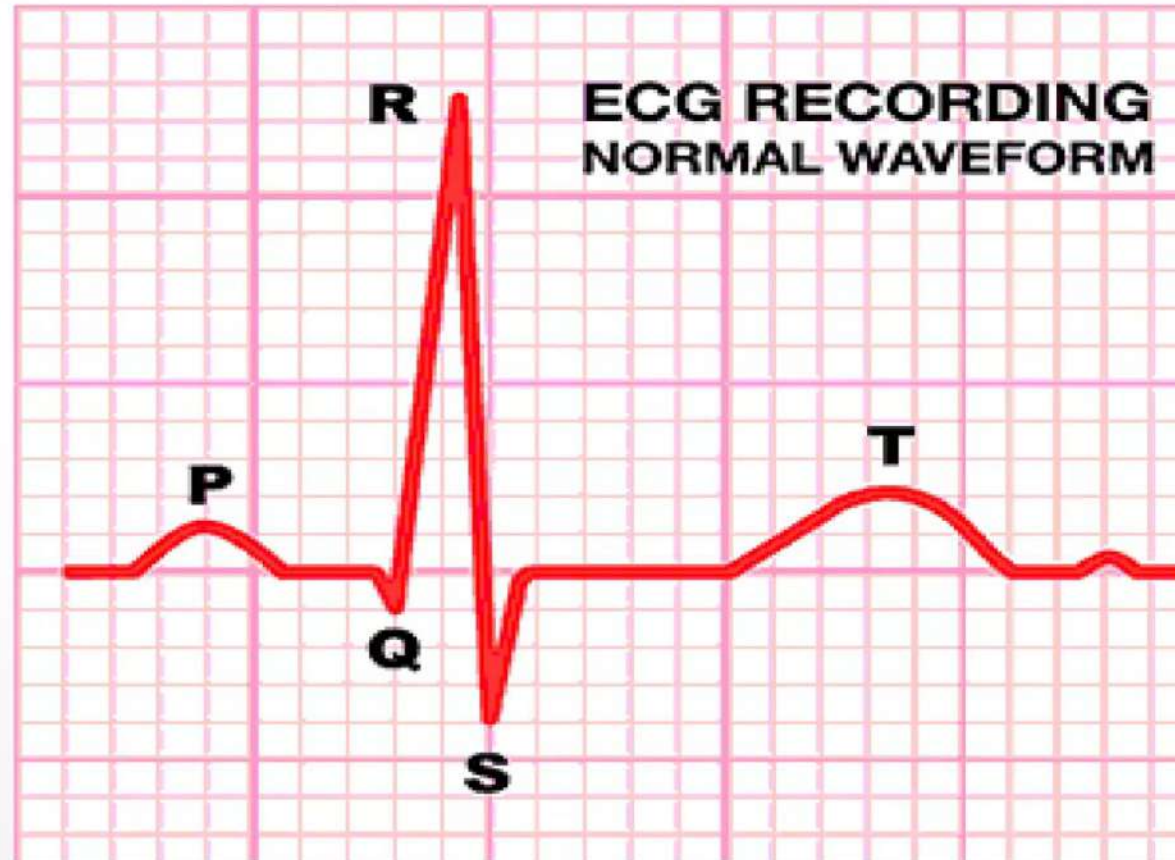
Heart Arrhythmia



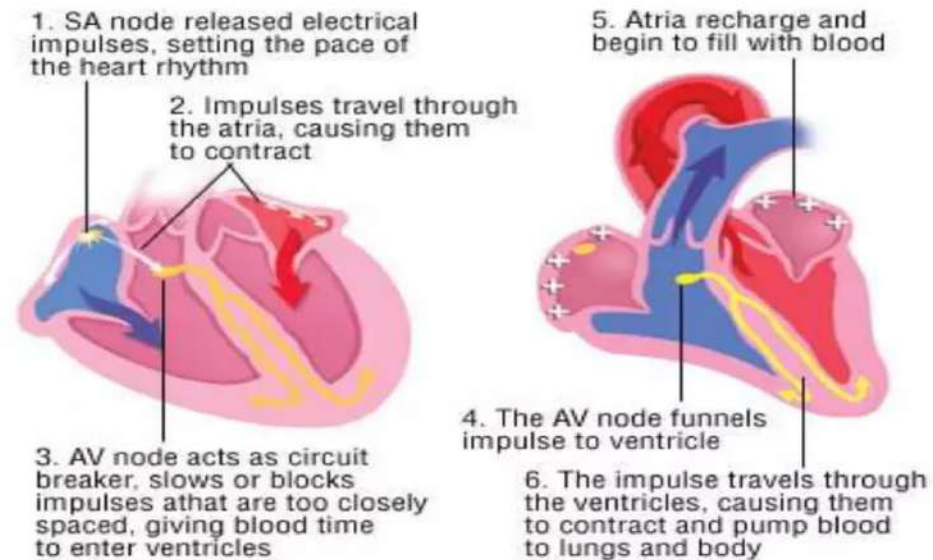
SWATILEKHA DAS
ASST PROFESSOR

ECG

- Normal PR interval- 0.12s to 0.2 s (3 to 5 small square)
- Normal QRS complex- $<0.12\text{s}$ (< 3 small squares)



- A cardiac arrhythmia is any abnormal heart rate or rhythm.
- In normal adults, the heart beats regularly at a rate of 60 to 100 times per minute.
- Under normal circumstances, the signal for a heartbeat comes from the heart's sinus node. It's the natural pacemaker located in the upper portion of the right atrium. From the sinus node, the heartbeat signal travels to the atrioventricular node or "A-V node," which is located between the atria. Next the signal travels through the bundle of His (pronounced HISS). It's made up of a series of modified heart muscle fibers located between the ventricles. The signal enters the muscles of the ventricles. This causes the ventricles to contract and produces a heartbeat.
- **Normal Cardiac rhythm :**
 - HR = 60-100 bpm
 - Sinus Origin
 - Normal Conducting Pathways
 - Normal Conducting Velocity



CLASSIFICATION:

1. **Sinus Arrhythmia** – originates from the sinoatrial node

- Sinus bradycardia - <60 bpm
- Sinus tachycardia- >100 bpm

2. **Atrial Arrhythmia** – originates from the atria

- Atrial tachycardia
- Atrial Flutter
- Atrial Fibrillation

3. **Ventricular Arrhythmia** – originates from the ventricle

- Ventricular premature beats
- Ventricular tachycardia
- Ventricular fibrillation



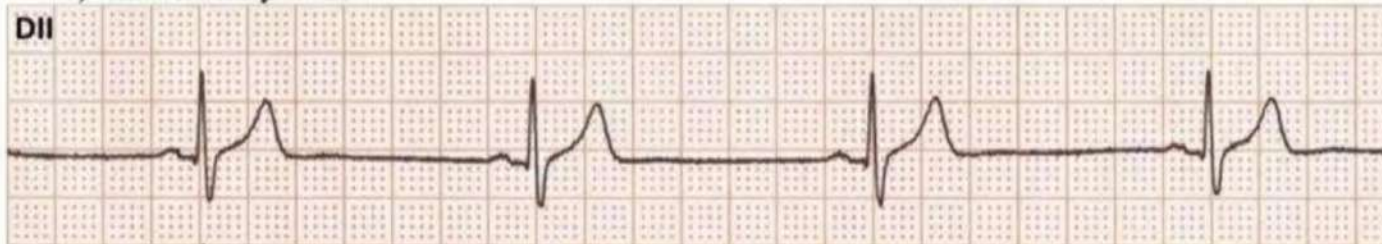
- **SINUS NODE DYSFUNCTIONS**

- This usually causes a slow heart rate (bradycardia), with a heart rate of 50 beats per minute or less. The most common cause is scar tissue that develops and eventually replaces the sinus node. Sinus node dysfunction also can be caused by coronary artery disease, hypothyroidism, severe liver disease, hypothermia, typhoid fever or other conditions. It also can be the result of vasovagal hypertonía, an unusually active vagus nerve.

A) Sinus tachycardia



B) Sinus bradycardia

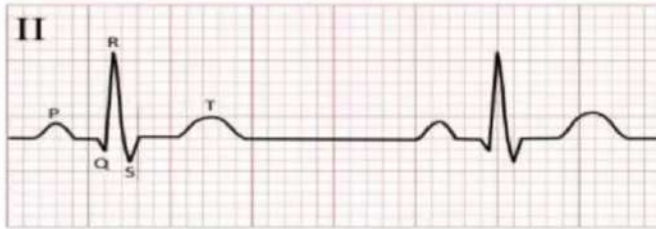


ATRIAL ARRHYTHMIAS

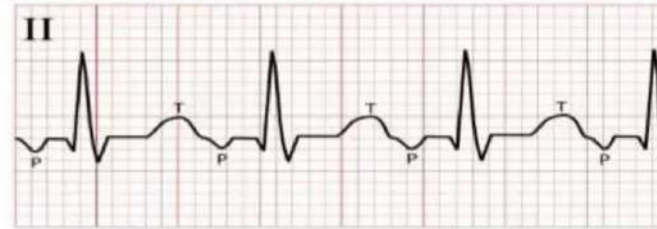
- **Atrial tachycardia** - During atrial tachycardia, an electrical impulse outside the sinus node fires repeatedly, often due to a short circuit — a circular electrical pathway. Electricity circles the atria again and again, causing the upper chambers to contract more than 100 times per minute. (A normal heart rate is between 60 and 100 beats per minute.) The rapid heart contractions prevent the chambers from filling completely between beats.
- An arrhythmia centered in the upper chambers of the heart is called a **supraventricular tachycardia** (SVT) — literally, fast "heartbeat above the ventricles" (lower chambers). Atrial tachycardia usually occurs for brief periods and starts and stops spontaneously. That's called **paroxysmal Atrial tachycardia**. If it continues, it is called **persistent atrial tachycardia**.
- **Atrial Flutter** - Atrial flutter results from an abnormal circuit inside the right atrium, or upper chamber of your heart. It beats extra fast, about 250-400 beats per minute. A normal heartbeat is 60-100 beats per minute.
- In atrial flutter, electrical impulses don't travel in a straight line from the top of your heart to the bottom. Instead, they move in a circle inside the upper chambers. As a result, your heart beats too fast, but still in a steady rhythm.
- **Atrial fibrillation** - This is a supraventricular arrhythmia that causes a rapid and irregular heartbeat, during which the atria quiver or "fibrillate" instead of beating normally. During atrial fibrillation, heartbeat signals begin in many different locations in the atria rather than in the sinus node.
- >350 beats per minute
- In AFib, the electrical signals that travel through the atria are fast and disorderly, which makes them quiver instead of squeezing strongly. This causes the heart to beat too fast and in a chaotic rhythm.
- The disordered heartbeat of atrial fibrillation cannot pump blood out of the heart efficiently. This causes blood to pool in the heart chambers and increases the risk of a blood clot forming inside the heart. The major risk factors for atrial fibrillation are age, high blood pressure, heart valve abnormalities, diabetes, and heart failure.



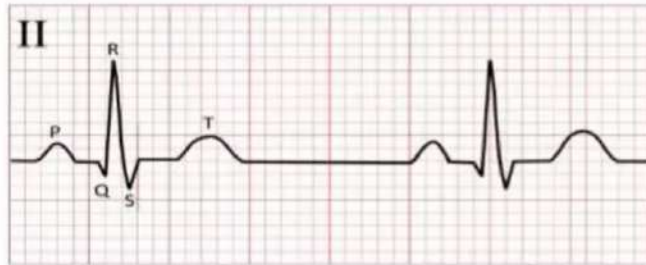
Sinus rhythm



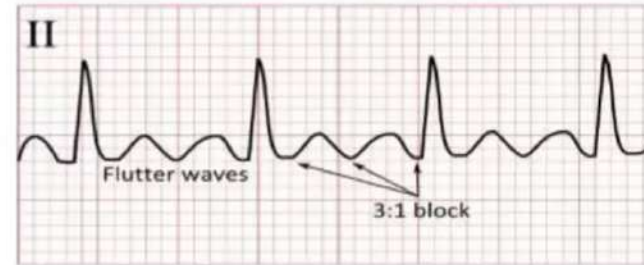
Atrial tachycardia



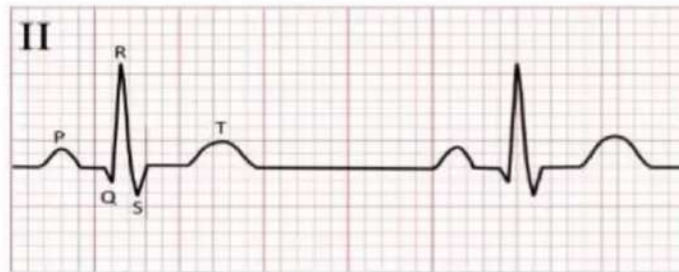
Sinus rhythm



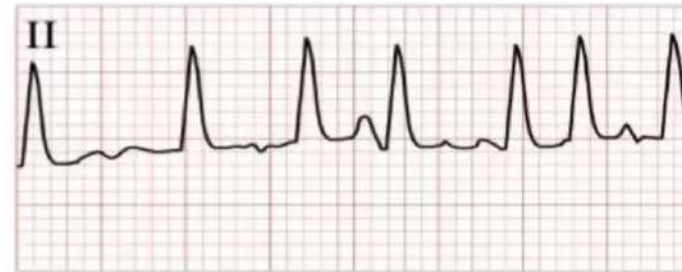
Atrial flutter



Sinus rhythm

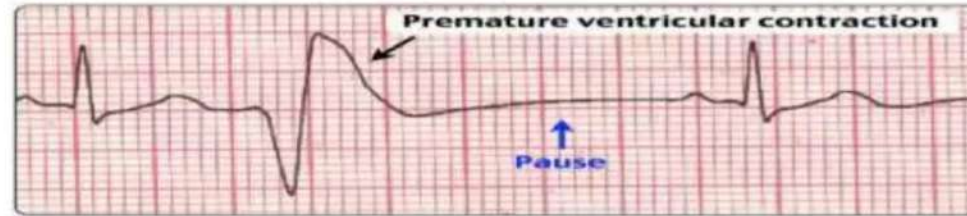


Atrial fibrillation



VENTRICULAR ARRHYTHMIAS:

- **Ventricular premature beats** - A ventricular premature beat is an extra heartbeat resulting from abnormal electrical activation originating in the ventricles (the lower chambers of the heart) before a normal heartbeat would occur.



- **Ventricular tachycardia (VT)** - This is an abnormal heart rhythm that begins in either the right or left ventricle. It may last for a few seconds (non-sustained VT) or for many minutes or even hours (sustained VT). Sustained VT is a dangerous rhythm and if it is not treated, it often progresses to ventricular fibrillation.

Ventricular Tachycardia



- **Ventricular fibrillation** - In this arrhythmia, the ventricles quiver ineffectively, producing no real heartbeat. The result is unconsciousness, with brain damage and death within minutes. Ventricular fibrillation is a cardiac emergency. Ventricular fibrillation can be caused by a heart attack, an electrical accident, a lightning strike or drowning.

Ventricular Fibrillation



- **HEART BLOCKS OR A- V BLOCKS:**

- In this type of arrhythmias, there is some problem conducting the heartbeat signal from the sinus node to the ventricles. There are three degrees of A-V block:

- First-degree A-V block, where the signal gets through, but may take longer than normal to travel from the sinus node to the ventricles
 - Continuously prolonged PR interval
 - Normal PR interval duration is 0.12s – 0.20s
 - Second-degree A-V block, in which some heartbeat signals are lost between the atria and ventricles
 - Mobitz type 1 (Wencheback) – progressively prolonged PR interval, then dropped QRS complex
 - Mobitz type 2 – Intermittent block, dropped QRS complex
 - Third-degree A-V block, in which no signals reach the ventricles, so the ventricles beat slowly on their own with no direction from above
 - Atria contracts independently of ventricles
 - No correlation between P waves and QRS complexes.

- Some causes of A-V block include cardiomyopathy, coronary artery disease, and medications such as beta blockers and digoxin.



First degree AV block



Second degree AV block (Mobitz I or Wenckebach)



Second degree AV block (Mobitz II)



Second degree AV block (2:1 block)



Third degree AV block with junctional escape



SYMPTOMS

- Symptoms of specific arrhythmias include:

- **Sinus node dysfunction** - There may not be any symptoms, or it may cause dizziness, fainting and extreme fatigue.
- **Supraventricular tachyarrhythmias** - These can cause palpitations (awareness of a rapid heartbeat), low blood pressure and fainting.
- **Atrial fibrillation** - Sometimes, there are no symptoms. This can cause palpitations; fainting; dizziness; weakness; shortness of breath; and angina, which is chest pain caused by a reduced blood supply to the heart muscle. Some people with atrial fibrillation alternate between the irregular heartbeat and long periods of completely normal heartbeats.
- **A-V block or heart block** - First-degree A-V block does not cause any symptoms. Second-degree A-V block causes an irregular pulse or slow pulse. Third-degree A-V block can cause a very slow heartbeat, dizziness and fainting.
- **VT** - Non-sustained VT may not cause any symptoms or cause a mild fluttering in the chest. Sustained VT usually causes lightheadedness or loss of consciousness and can be lethal.
- **Ventricular fibrillation** - This causes absent pulse, unconsciousness and death.



DIAGNOSIS

- **History taking**
- Duration of illness & triggering factor
- Relationship to exercise, meals, stress
- Medical history- H/O of tachycardia, cardiac problems
- Prior medications, toxins
- H/O allergies
- Sudden death in the family, cardiac disease
- **Physical examination**
- **Investigations**
- 12 lead ECG, cardiac rhythm strip, CBC
- X-ray chest, echocardiography
- Electrolyte- potassium, glucose, calcium, magnesium
- Toxicology screen
- Arterial blood gases
- 24 hrs ambulatory Holter monitoring
- Electro physiological studies



TREATMENT

- The treatment of a cardiac arrhythmia depends on its cause:

- **Sinus node dysfunction** - In people with frequent, severe symptoms, the usual treatment is a permanent pacemaker.
- **Supraventricular tachyarrhythmias** - The specific treatment depends on the cause of the arrhythmia. In some people, massaging the carotid sinus in the neck will stop the problem. Other people need medications such as beta-blockers, calcium channel blockers, digoxin (Lanoxin) and amiodarone (Cordarone). Some patients respond only to a procedure called radiofrequency catheter ablation, which destroys an area of tissue in the A-V node to prevent excess electrical impulses from being passed from the atria to the ventricles.
- **Atrial fibrillation** - Atrial fibrillation resulting from an overactive thyroid can be treated with medications or surgery. Fibrillation resulting from mitral or aortic valve disease may be treated by replacing damaged heart valves. Medications, such as beta-blockers (for example atenolol and metoprolol), amiodarone, diltiazem (Cardizem, Tiazac) or verapamil (Calan, Isoptin, Verelan), can be used to slow the heart rate. Drugs such as amiodarone can be used to reduce the chances that the atrial fibrillation will return. Other treatment options include radiofrequency catheter ablation, or electrical cardioversion, a procedure that delivers a timed electrical shock to the heart to restore normal heart rhythm.
- **A-V block** - First-degree A-V block typically does not require any treatment. People with second-degree A-V block may be monitored with frequent EKGs, especially if they do not have any symptoms and have a heart rate that is adequate for their daily activities. Some patients with second-degree heart block may require permanent pacemakers. Third-degree A-V block is almost always treated with a permanent pacemaker.
- **VT** - Non-sustained VT may not need to be treated if there is no structural damage to the heart. Sustained VT always needs treatment, either with intravenous medication or emergency electrical shock (defibrillation), which can restore the heart's normal rhythm.
- **Ventricular fibrillation** - This is treated with defibrillation, giving the heart a measured electrical shock to restore normal rhythm. The electrical shock can be delivered on the skin over the heart in an emergency situation. People who have survived ventricular fibrillation and those at high risk are potential candidates for an automatic implantable cardioverter defibrillator. The device is similar to a pacemaker, with wires attached to the heart that connect an energy source placed under the skin. The procedure is done in the operating room.



Exercise prescription with a Pacemaker or Implantable Cardioverter Defibrillator

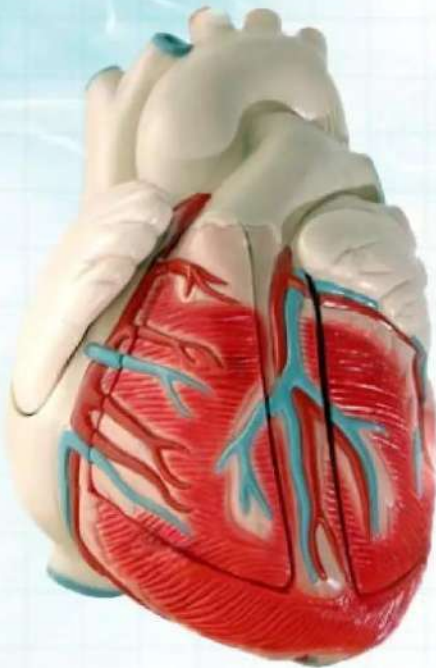
- Exercise plays an important role in rehabilitation after implantation of a pacemaker or other cardioverter defibrillator because it will counteract the deconditioning that occurred prior to implantation, and reduce your risk factors for heart disease, including high blood pressure and cholesterol, diabetes and obesity. The key to maximizing the benefits of exercise is to follow a well-designed program that you can stick to over the long-term.
- **Aerobic Activity**-Aerobic activity increases your heart rate and breathing. Build up to doing at least 150 minutes/week of moderate-intensity activity (such as a brisk walk, light cycling or water exercise). You'll improve your stamina and heart health
- Activity- Any rhythmic, continuous activity
- Frequency- 3-5 days a week
- Intensity - Fairly light to somewhat hard
- Duration- Start w/5-10 mins. Gradually build up to 20-60 minutes.
- **Strength Training** - Strength training can help your muscles work better and increases your strength for daily and recreational activities.
- Activity - Hand weights, resistance bands, weight machines, or your own body; for example, kitchen counter push-ups or chair sit-to-stands.
- Frequency - 2-3 days/week. Rest day in between
- Intensity- Start with light effort. Build up to medium effort.
- Duration - 10-15 repetitions (for each major muscle group). Perform 1-3 times.



Cardiac arrhythmias

Arrhythmia Formation

- Arrhythmias can arise from problems in the:
 - Sinus node
 - Atrial cells
 - AV junction
 - Ventricular cells
- Arrhythmias caused by 2 main mechanism:
 - Automaticity
 - Re-entry.



Cardiac arrhythmias

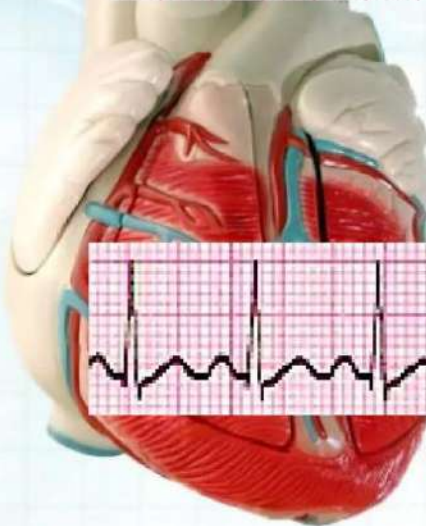
SA Node Problems

The SA Node can:

- fire too slow *Sinus Bradycardia*



- fire too fast *Sinus Tachycardia*



Ventricular Cell Problems

Cardiac arrhythmias

Ventricular cells can:
fire occasionally from
1 or more foci

**Premature
Ventricular
Contractions
(PVCs)**



- fire continuously from multiple foci

**Ventricular
Fibrillation**



- fire continuously due to a looping re-entrant circuit

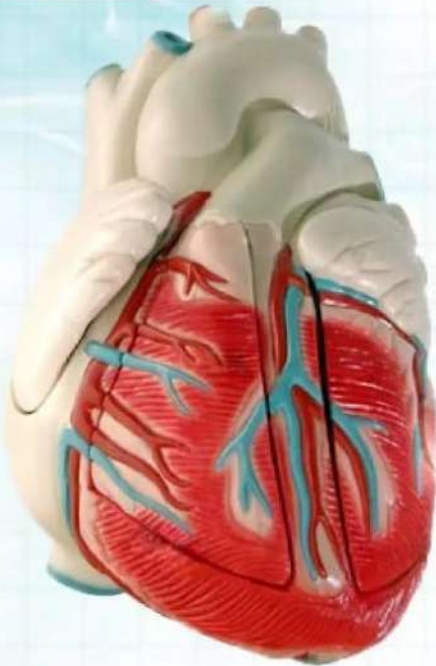
**Ventricular
Tachycardia**



Sinus Bradycardia



Cardiac arrhythmias

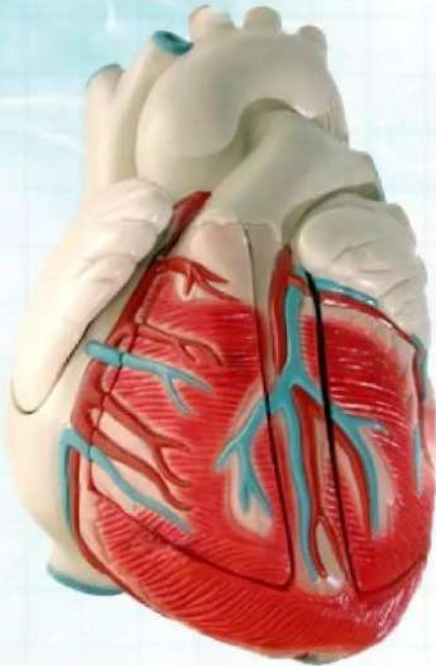


- A sinus rate of less than 60 b.p.m. during the day or less than 50 .m. at night is known as sinus bradycardia.
- CAUSES

Physiological

- ❖ In athletes
- ❖ During sleep

Cardiac arrhythmias

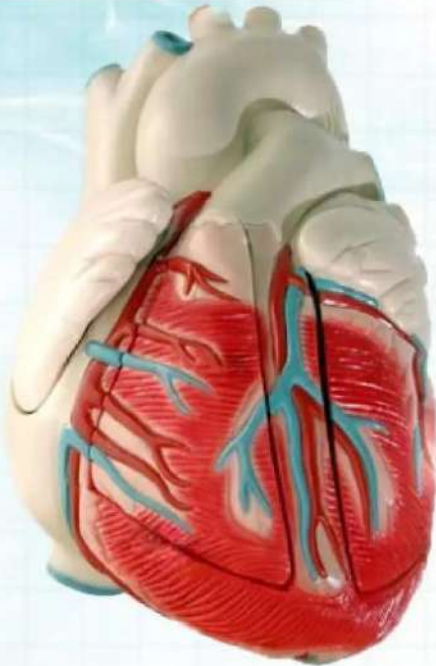


Pathological

Extrinsic causes

1. Hypothermia, hypothyroidism, cholestatic jaundice and raised intracranial pressure.
2. Drug therapy with beta-blockers, digitalis and other antiarrhythmic drugs.
3. Neurally mediated syndromes

Cardiac arrhythmias



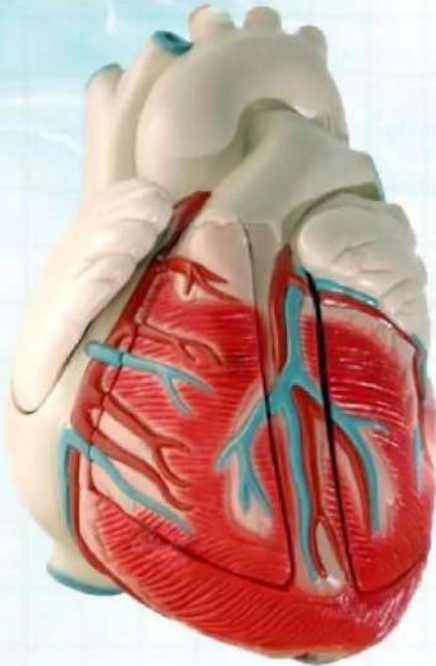
Intrinsic causes

1. Acute ischaemia and infarction of the sinus node (as a complication of acute myocardial infarction).
2. Chronic degenerative changes such as fibrosis of the atrium and sinus node (sick sinus syndrome).

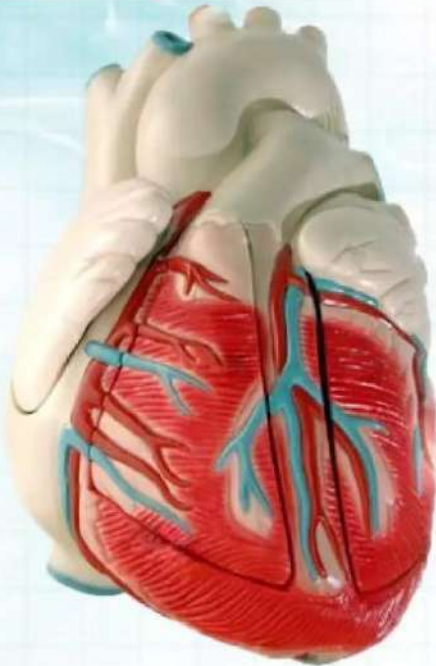
Sick sinus syndrome

Sick sinus syndrome or *sinoatrial disease* is usually caused by idiopathic fibrosis of the sinus node. Other causes of fibrosis such as ischemic heart disease, cardiomyopathy or myocarditis can also cause the syndrome. Patients present episodes of sinus bradycardia, sinus arrest, paroxysmal supraventricular tachycardia or tachy-brady syndrome.

**Cardiac
arrhythmias**



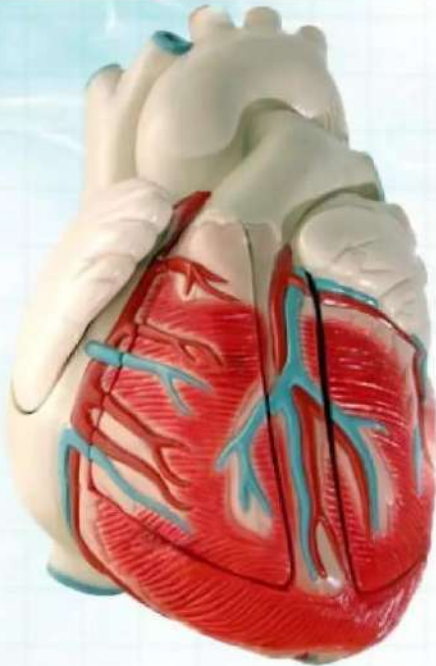
Cardiac arrhythmias



CLINICAL FEATURES

- Pt may be asymptomatic
- Severe bradycardia may cause weakness, confusion & syncope
- Atrial & ventricular ectopics are more apparent to occur with slow sinus rate.

Cardiac arrhythmias



TREATMENT

- Identification & treatment of the cause
- Acute symptomatic bradycardia responds to atropine 0.6 mg i.v.
- Temporary pacemaker in symptomatic bradycardia if there is reversible cause
- Permanent pacemaker in symptomatic irreversible cause

Cardiac arrhythmias

Sinus tachycardia

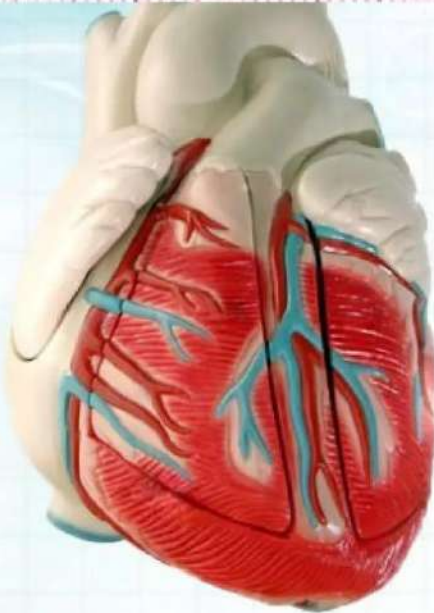
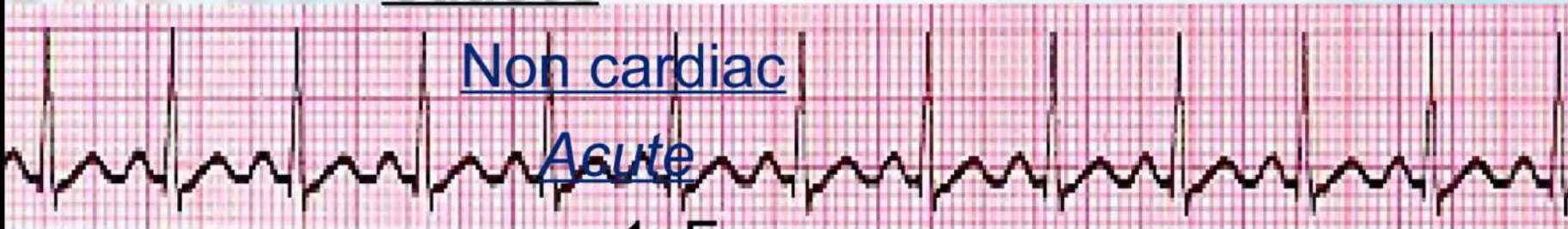
- Resting sinus rate of more than 100 b.p.m. is known as sinus tachycardia.

Causes

Non cardiac

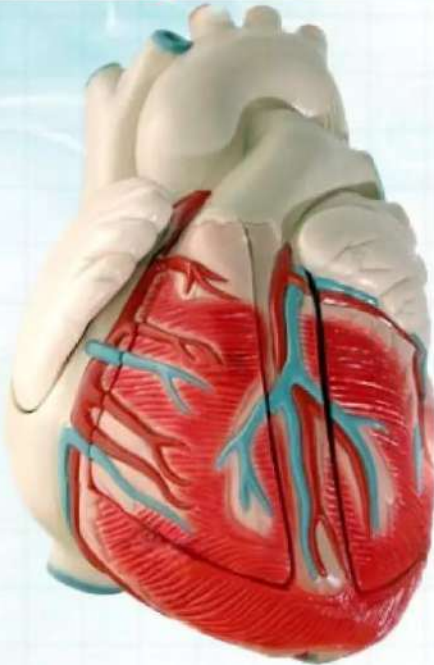
Acute

1. Fever
2. Hypovolemia
3. Pain
4. Infection
5. Exercise
6. Emotion/anxiety



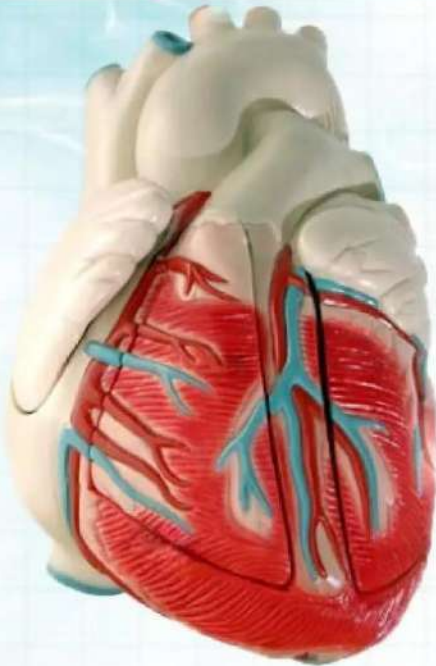
supraventricular tachycardia (SVT)

Cardiac arrhythmias



- This is a tachycardia occurring in episodes with rate of 140-240, as a result of re-entry or rapidly firing ectopic focus in the atria or AV node. It may last from few seconds to many hours (if untreated)
- Predisposing factors:
 - Anxiety
 - Excess tobacco or coffee
 - Hyperthyroidism
 - Exertion
 - Alcohol

Cardiac arrhythmias



vagotonic maneuvers

- Carotid sinus message
- Valsalva maneuver
- facial immersion in cold water
- Pressure on eyes
- Self inducing vomiting
- Breath holding
- Lowering the head between the knees

Automatic external defibrillator



THANK YOU

